

HOLLISTER INCORPORATED

Digital Manufacturing Footprint

Solution Architecture & Baseline Configuration

From Manufacturing Line to Enterprise Supply Chain

Baseline Plant: Hollister Ballina, Co. Mayo, Ireland

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1. Executive Summary

This Solution Architecture defines the complete digital manufacturing footprint for Hollister Incorporated, establishing a modular, scalable, and cGMP-compliant baseline configuration starting at the Ballina, Ireland plant. The architecture spans from individual sensor endpoints on manufacturing equipment to enterprise-wide supply chain integration, designed with the explicit goal of replicability across all Hollister global manufacturing facilities.

The design addresses six fundamental requirements: modularity (plug-and-play component architecture enabling graceful change introduction), scalability (from single manufacturing line to full enterprise), resilience (fault tolerance at every layer with no single points of failure), security (defense-in-depth aligned with IEC 62443 and FDA 21 CFR Part 11), precision (metrology-grade measurement traceability), and compliance (end-to-end V&V supporting IQ/OQ/PQ qualification and cGMP requirements).

A critical design constraint is the accommodation of an interim PLM solution that provides immediate value while maintaining clean migration paths to a permanent PLM platform when the organization is ready. Similarly, the architecture supports Manufacturing on Demand (MOD) through configurable production recipes, dynamic resource allocation, and rapid changeover capabilities.

Baseline Scope: One manufacturing line at the Ballina facility, fully instrumented and integrated, serving as the validated template for horizontal and vertical scaling across all Hollister plants globally.

2. Design Principles & Architectural Framework

2.1 Governing Principles

Every design decision in this architecture is evaluated against the following non-negotiable principles, which reflect both operational excellence requirements and regulatory obligations inherent to medical device manufacturing.

Principle	Description	Compliance Mapping
Modularity	Each component (sensor, PLC, DCS module, compute node) is independently deployable, replaceable, and upgradeable without disrupting adjacent systems.	IEC 62443 Zone/Conduit, ISA-95 Levels
Scalability	Architecture scales horizontally (add lines, plants) and vertically (add analytics, AI layers) without re-architecture.	ISA-95 Enterprise Integration
Resilience	No single point of failure at any layer. Graceful degradation with automatic failover. Local autonomy preserved during WAN outages.	IEC 61508 SIL, FDA 21 CFR 211
Security	Defense-in-depth: network segmentation, zero-trust, encrypted communications, role-based access, audit trails.	IEC 62443, FDA 21 CFR Part 11, GDPR
Precision	Metrology-grade measurement with traceability to international standards. Calibration management integrated into operations.	ISO 17025, FDA 21 CFR Part 211.68
Compliance-by-Design	cGMP compliance, V&V, IQ/OQ/PQ built into every layer rather than bolted on.	FDA 21 CFR Parts 11/210/211/820, EU MDR Annex XIII
Change Control	Graceful introduction of change with full impact analysis, rollback capability, and validated state management.	GAMP 5, ICH Q10
Interoperability	Open standards preferred (OPC-UA, MQTT, ISA-95 B2MML) to avoid vendor lock-in and enable ecosystem evolution.	ISA-95, IEC 62264, OPC-UA

2.2 ISA-95 / Purdue Model Alignment

The architecture strictly follows the ISA-95 (Purdue) model for hierarchical separation of concerns. This is not merely a network segmentation strategy but a fundamental organizational principle governing data flow, latency requirements, and security boundaries.

Level	Function	Hollister Implementation	Latency Req.
Level 0	Physical Process (Sensors/Actuators)	Field instruments, temperature/pressure/flow sensors, vision systems, actuators	<1 ms

Level 1	Basic Control (PLCs/Safety)	Allen-Bradley/Siemens PLCs, safety controllers, local HMIs	1-10 ms
Level 2	Area Supervisory (DCS/SCADA)	DCS per manufacturing line, SCADA for facility-wide monitoring, historian	100 ms-1 s
Level 3	Site Operations (MES/MOM)	Manufacturing Execution System, batch management, quality management	1-30 s
Level 3.5	DMZ / Industrial DMZ	Data diodes, protocol break, OPC-UA aggregation servers	N/A (security)
Level 4	Enterprise (ERP/PLM/SCM)	SAP/Oracle ERP, interim PLM, supply chain management, analytics	Minutes-hours
Level 5	Enterprise Network/Cloud	Multi-cloud analytics, digital twin, enterprise data lake	Variable

2.3 Hierarchical Scope Model

The digital footprint is organized hierarchically to enable modular deployment and independent qualification:

- Manufacturing Station: Individual machine or work cell with its associated sensors, PLCs, and local compute
- Manufacturing Line: Collection of stations forming a complete production process for a product family
- Shop Floor: All manufacturing lines within a production area sharing common utilities and infrastructure
- Plant: Complete facility including all shop floors, warehousing, utilities, and site infrastructure
- Enterprise: All plants connected via supply chain integration, shared analytics, and unified governance

Each level in this hierarchy has its own qualification package (IQ/OQ/PQ), change control boundary, and data sovereignty model. The baseline configuration defined herein covers Levels 0-3 for a single manufacturing line, with integration points defined for Levels 3.5-5.

3. Baseline Plant Configuration: Ballina, Ireland

3.1 Plant Profile & Assumptions

The Hollister Ballina facility in Co. Mayo, Ireland serves as the baseline plant for this digital footprint. As a medical device manufacturing facility operating under both FDA and EU MDR regulatory frameworks, it represents the most stringent compliance requirements, making it an ideal baseline that will satisfy requirements for any subsequent plant deployment.

Parameter	Baseline Assumption
Facility Type	Medical device manufacturing (ostomy, continence care, wound care products)
Regulatory Framework	FDA 21 CFR 210/211/820, EU MDR 2017/745, ISO 13485:2016
Number of Manufacturing Lines (Initial)	6-8 production lines (baseline config for 1 line)
Clean Room Classes	ISO 7 (Class 10,000) and ISO 8 (Class 100,000) per ISO 14644
Operating Schedule	24/7 continuous operations, 3 shifts
Environmental Zones	Controlled (clean rooms), Semi-controlled (packaging), Uncontrolled (warehouse/utilities)
Existing Infrastructure	Legacy PLCs (mix of vendors), limited historian, manual quality data collection
Network Backbone	Existing Ethernet to be upgraded to segmented industrial Ethernet

3.2 Baseline Manufacturing Line Architecture

A single manufacturing line serves as the repeatable unit of deployment. The baseline line is assumed to consist of the following stations (adapted to Hollister product families):

Station	Function	Key Equipment	Sensor Density
S1: Raw Material Prep	Material handling, blending, pre-treatment	Mixers, dispensers, material handling robots	Medium (12-18 sensors)
S2: Forming/Molding	Primary forming of product components	Injection molders, thermoformers, extruders	High (25-40 sensors)
S3: Assembly	Component assembly and sub-assembly	Pick-and-place, ultrasonic welders, adhesive dispensers	High (20-35 sensors)
S4: Coating/Treatment	Surface treatment, adhesive application	Coating systems, UV cure, drying ovens	High (20-30 sensors)
S5: Inspection/Testing	In-line quality inspection and functional testing	Vision systems, leak testers, force testers	Very High (30-50 sensors)

S6: Packaging	Primary and secondary packaging, labeling	Pouch sealers, cartoners, labelers, case packers	Medium (15-25 sensors)
S7: Palletizing/Shipping	Tertiary packaging, palletizing, shipping prep	Palletizers, stretch wrappers, labeling	Low (8-12 sensors)

4. Programmable Logic Controller (PLC) Configuration

4.1 PLC Strategy & Vendor Recommendation

The PLC strategy follows a standardized-platform approach with a single primary vendor ecosystem to minimize integration complexity, spare parts inventory, and training requirements while maintaining the flexibility to integrate specialized controllers where process-specific requirements demand it.

4.1.1 Primary PLC Platform: Rockwell Automation Allen-Bradley

Recommendation: Rockwell Automation ControlLogix 5580 family as the primary PLC platform, with CompactLogix 5380 for smaller stations and GuardLogix for safety-rated applications.

Rationale for Rockwell selection in a Hollister context:

- Market-leading installed base in North American and European medical device manufacturing
- Integrated safety (GuardLogix) eliminates separate safety controller hardware for most applications
- Native EtherNet/IP with CIP (Common Industrial Protocol) provides unified communications
- FactoryTalk suite provides SCADA/historian/analytics within a single ecosystem
- Strong regulatory validation support with established 21 CFR Part 11 compliance templates
- Global service network including Ireland-based field engineering

4.1.2 Secondary PLC Platform: Siemens (Interoperability Layer)

Siemens S7-1500 series is recommended as a secondary platform for specific use cases: legacy integration where Siemens PLCs are already deployed, specialized motion control applications where Siemens SIMOTION excels, and interoperability validation to ensure the architecture is not single-vendor dependent.

4.2 PLC Allocation per Manufacturing Line

Each manufacturing line requires a carefully balanced allocation of PLCs that balances granularity of control against complexity of integration. The following allocation is recommended per baseline manufacturing line:

Station	Primary PLC	Qty	Safety PLC	Qty	I/O Points (est.)	Justification
S1: Raw Material Prep	CompactLogix 5380	1	Compact GuardLogix	1	64-96	Lower complexity; compact form factor sufficient

S2: Forming/Molding	ControlLogix 5580	2	GuardLogix 5580	1	192-256	High I/O density; critical process control; redundant pair recommended
S3: Assembly	ControlLogix 5580	1	GuardLogix 5580	1	128-192	Complex motion coordination; integrated safety zones
S4: Coating/Treatment	ControlLogix 5580	1	Compact GuardLogix	1	96-160	Precision process control; thermal management
S5: Inspection/Testing	CompactLogix 5380	2	N/A (non-safety)	0	64-128	Vision system coordination; parallel test stations
S6: Packaging	ControlLogix 5580	1	GuardLogix 5580	1	128-192	High-speed coordination; serialization/track-and-trace
S7: Palletizing	CompactLogix 5380	1	Compact GuardLogix	1	48-64	Robotic coordination; lower complexity
Line Coordinator	ControlLogix 5580	1	N/A	0	32-48	Line-level orchestration; recipe management; MES interface

Total per Manufacturing Line: 10 primary PLCs + 6 safety PLCs = 16 controllers. Estimated total I/O: 752-1,136 points.

4.3 PLC Configuration per Plant (Full Deployment)

Scaling the per-line allocation to a full plant with 6-8 manufacturing lines:

Component	Per Line	6 Lines	8 Lines	Plant Infrastructure	Plant Total (8 Lines)
ControlLogix 5580	6	36	48	4 (utilities, BMS, WH)	52
CompactLogix 5380	4	24	32	6 (aux systems)	38
GuardLogix 5580	3	18	24	2 (plant safety)	26
Compact GuardLogix	3	18	24	2 (plant safety)	26
Total Controllers	16	96	128	14	142
Est. I/O Points	~950	~5,700	~7,600	~1,200	~8,800

4.4 PLC Network Architecture

All PLCs communicate via EtherNet/IP on a segmented, ring-topology industrial Ethernet network. Each manufacturing line operates on its own VLAN with a managed switch stack. The network design follows:

- Device Level Ring (DLR): EtherNet/IP ring topology per station for sub-millisecond failover
- Line Level Backbone: Managed Stratix 5700 switches per line with inter-line trunking
- Plant Backbone: Stratix 5400 Layer 3 switches for routing between line VLANs and plant DMZ
- Redundancy: All critical PLCs deployed in 1oo2 (one-out-of-two) redundant pairs for forming/molding stations
- Time Synchronization: IEEE 1588 Precision Time Protocol (PTP) for sub-microsecond synchronization across all PLCs

4.5 PLC Programming Standards

All PLC programs shall adhere to IEC 61131-3 with the following Hollister-specific standards:

- Structured Text (ST) as the primary language for complex logic and calculations
- Function Block Diagram (FBD) for regulatory logic and interlocks
- Sequential Function Charts (SFC) for batch/recipe control aligned with ISA-88
- Ladder Logic (LD) reserved only for discrete I/O interlocking where required by maintenance familiarity
- All programs version-controlled in a validated source control system (e.g., Rockwell AssetCentre or AVEVA Teamwork)
- Add-On Instructions (AOIs) library for standardized functions reusable across all plants

5. Distributed Control System (DCS) Configuration

5.1 DCS Strategy

For Hollister's medical device manufacturing, the architecture recommends a hybrid DCS/PLC approach rather than a pure DCS deployment. This reflects the discrete manufacturing nature of Hollister's products (as opposed to continuous process industries) while providing the supervisory coordination, batch management, and regulatory data management capabilities that a DCS excels at.

5.1.1 Recommended Platform: Rockwell PlantPAx

Primary Recommendation: Rockwell PlantPAx 5.0 as the DCS layer, providing a seamless integration with the ControlLogix PLC infrastructure. PlantPAx is implemented as a distributed control architecture on top of the existing PLC hardware, eliminating the need for separate DCS hardware while providing full DCS functionality.

Alternative (for consideration): Emerson DeltaV or Siemens PCS 7 if the organization requires a traditional DCS with independent hardware. This would add cost and integration complexity but provides vendor diversity for critical processes.

5.2 DCS Scope per Manufacturing Line

The DCS layer provides the following functions per manufacturing line:

DCS Function	Implementation	Components	Qualification
Supervisory Control	PlantPAx process objects running on Line Coordinator PLC	Recipe manager, sequence coordinator, material tracker	IQ/OQ/PQ per ISA-88
Batch Management	ISA-88 compliant batch engine	Recipe phases, unit procedures, master recipes	GAMP 5 Cat 4/5
Operator Interface	FactoryTalk View SE distributed HMI	3x operator stations per line, 1x supervisor station	21 CFR Part 11 validated
Historian	FactoryTalk Historian SE	Continuous data at 1-second resolution; event-driven for alarms	Validated, Part 11 compliant
Alarm Management	PlantPAx alarm strategy per ISA-18.2	Alarm rationalization, shelving, suppression audit trail	IQ/OQ required
Report Generation	FactoryTalk VantagePoint + SSRS	Batch reports, OEE, quality metrics, environmental logs	Validated output

5.3 DCS Architecture per Plant

At the plant level, the DCS provides coordination across all manufacturing lines plus shared utilities:

- Plant-Wide Historian: Consolidated data from all lines with 5+ year retention for trend analysis and regulatory queries
- Centralized Alarm Console: Aggregated alarm view for plant operations center, conforming to EEMUA 191
- Utility Systems Integration: BMS (HVAC, clean room), purified water systems, compressed air, electrical distribution
- Environmental Monitoring System (EMS): Integrated particle counters, differential pressure, temperature, humidity across all clean rooms
- MES Gateway: OPC-UA server providing standardized data access for Level 3 systems

5.4 DCS Redundancy & High Availability

The DCS layer is configured for high availability with the following redundancy model:

- Server Redundancy: Hot standby for all PlantPax servers (application, historian, HMI) with automatic failover < 10 seconds
- Network Redundancy: Dual-homed Ethernet connections with RSTP for all DCS nodes
- Historian Redundancy: Primary + mirror historian with store-and-forward buffering at each data source
- Operator Station Redundancy: Any operator station can control any line (role-based, not hardware-based)
- UPS Protection: All DCS components on dedicated UPS with minimum 30-minute runtime plus generator backup

6. Comprehensive Sensor Architecture

6.1 Sensor Strategy Overview

The sensor architecture is organized into three categories: vendor-specific sensors that are tied to particular equipment, common sensors deployed uniformly across all plants, and facility-level sensors for building and environmental management. All sensors are selected for Industry 4.0 readiness with native digital communication (IO-Link, EtherNet/IP, HART, or PROFINET) to enable direct data access without analog signal degradation.

6.2 Manufacturing Equipment Sensors (Vendor-Specific)

These sensors are selected based on specific machine types and process requirements. The following represents the recommended sensor suite per station type:

6.2.1 Station S2: Forming/Molding — Highest Sensor Density

Measurement	Sensor Type	Recommended Vendor/Model	Qty per Station	Protocol	Accuracy
Mold Temperature	RTD (Pt100) multi-zone	Endress+Hauser iTEMP TMT72	8-12	HART/IO-Link	±0.1°C
Injection Pressure	Piezoelectric cavity pressure	Kistler 6183C	4-6	ComoNeo analog + digital	±0.5% FS
Clamping Force	Strain gauge load cell	HBM C9B	2-4	Analog/amplifier	±0.02% FS
Cycle Time	Inductive proximity	Balluff BES M18	4-6	IO-Link	±0.1 ms
Material Flow	Coriolis mass flow	Endress+Hauser Promass F300	1-2	EtherNet/IP	±0.1%
Barrel Temperature	Thermocouple Type J/K	WatlowDERA	4-8	HART	±1.5°C
Hydraulic Pressure	Pressure transmitter	WIKA S-20	4-6	IO-Link	±0.25% FS
Part Presence	Photoelectric	SICK W16	6-8	IO-Link	N/A (discrete)
Part Dimensions (In-line)	Laser micrometer	Keyence LS-9030	2-4	EtherNet/IP	±1 µm

6.2.2 Station S5: Inspection/Testing — Highest Precision

Measurement	Sensor Type	Recommended Vendor/Model	Qty per Station	Protocol	Accuracy
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Visual Inspection	Machine vision camera	Cognex In-Sight 9912	4-8	GigE Vision/EtherNet/IP	Sub-pixel
Surface Defect	3D line scan profiler	Keyence LJ-X8000	2-4	EtherNet/IP	±0.5 µm Z
Leak Testing	Pressure decay tester	ATEQ F620	2-4	Serial/EtherNet	±0.01 mbar
Force/Peel Testing	Load cell with linear actuator	Mark-10 M5-100	2-4	USB/Serial	±0.1% FS
Dimensional (CMM proxy)	Laser triangulation	Micro-Epsilon scanCONTROL	1-2	EtherNet/IP	±2 µm
Color/Appearance	Spectrophotometer inline	X-Rite Ci7860	1-2	Serial/EtherNet	ΔE* < 0.04
Weight Check	Precision checkweigher	Mettler Toledo XS	1-2	EtherNet/IP	±0.01 g
Barcode/2D Code	Vision reader	Cognex DataMan 475V	2-4	EtherNet/IP	99.9% read rate

6.3 Common Sensors (Standardized Across All Plants)

The following sensors are deployed identically across every plant and manufacturing line, enabling cross-plant comparisons, unified analytics, and simplified spare parts management.

6.3.1 Environmental Monitoring (Clean Room & Controlled Areas)

Measurement	Sensor	Vendor/Model	Deployment	Protocol
Particle Count (0.5-5 µm)	Optical particle counter	TSI AeroTrak 9310	1 per clean room zone + critical points	Modbus TCP/MQTT
Temperature	RTD (Pt100) with transmitter	Vaisala TMW110	Grid: 1 per 50 m² in clean rooms	Modbus TCP
Relative Humidity	Capacitive humidity/temp	Vaisala HMT330	Grid: 1 per 100 m²	Modbus TCP
Differential Pressure	Low-range DP transmitter	Setra Model 267	Every room/corridor boundary	BACnet/Modbus
Airflow Velocity	Thermal anemometer	TSI VelociCalc 9565	LAF hoods, critical air returns	Analog/logged
CO2 / VOC	NDIR CO2 + PID VOC	Honeywell AQ110 series	Occupied areas, solvent zones	BACnet

6.3.2 Utility Monitoring

Utility	Measurements	Sensor Vendor/Model	Deployment
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Compressed Air	Pressure, flow, dew point, oil content	BEKO METPOINT series (OCV, DPM, FLM)	Main header + each line drop
Purified Water (WFI/PW)	Conductivity, TOC, temperature, flow	Endress+Hauser Liquiline CM442 + Memosens	Generation, storage, each use point
Electrical Power	V, I, kW, kWh, PF, harmonics	Schneider PowerLogic PM8000	Main switchgear + each MCC/panel
HVAC/AHU	Supply/return temp, filter DP, damper pos	Siemens Desigo CC sensors	Each AHU serving clean rooms
Steam/Hot Water	Pressure, temperature, flow	Endress+Hauser Prowirl F200 + TMT82	Boiler headers, sterilization points
Nitrogen/Specialty Gas	Pressure, purity (O2 content), flow	Hamilton VisiFerm (purity) + E+H flow	Generation/storage + use points

6.3.3 Machine Health / Predictive Maintenance

Measurement	Sensor	Vendor/Model	Deployment
Vibration (tri-axial)	MEMS accelerometer + ICP piezo	Fluke 3561 FC (wireless) + PCB 356A45 (wired)	Every rotating machine >5 kW
Motor Current Signature	Current transformer + analyzer	Schneider PowerLogic ION9000	Critical motors on forming/packaging
Bearing Temperature	RTD skin sensor	ifm TN2531 (IO-Link)	Critical bearing housings
Oil Condition	Dielectric / moisture / particle	HYDAC Lab CM-A5/2	Hydraulic systems on presses
Ultrasonic Leak Detection	Airborne ultrasonic	SDT 340 (portable + fixed)	Compressed air leaks, steam traps

6.4 Camera Systems

Camera deployments serve three distinct purposes: machine vision (quality), process monitoring, and facility security. Each has different specifications and governance requirements.

6.4.1 Machine Vision Cameras (Quality / Inspection)

Covered under Station S5 above. These are high-resolution, precisely calibrated cameras treated as metrology instruments with full IQ/OQ/PQ qualification. Recommended: Cognex In-Sight 9000 series and Keyence CV-X400 series.

6.4.2 Process Monitoring Cameras

Application	Camera Type	Vendor/Model	Resolution	Deployment
Operator Activity (ergonomics/safety)	Fixed IP camera with analytics	Axis P3267-LVE with ACAP	5 MP	2 per station (overhead + side)
Material Flow Tracking	Fixed wide-angle	Axis M3068-P (fisheye)	12 MP	Corridor intersections, staging areas
Changeover Verification	PTZ with preset tours	Axis Q6135-LE	1080p/60 fps	1 per line (critical changeover area)
Thermal Monitoring	Thermal overlay	FLIR A500/A700	640x480 thermal	Electrical panels, high-heat processes

6.4.3 Facility Security Cameras

Zone	Camera Type	Vendor/Model	Deployment Density
Perimeter / Parking	4K bullet with IR	Axis Q1808-LE	Per security assessment (typically 1/25m perimeter)
Building Entrances	Dome with face detection	Axis P3268-LVE	Every entrance/exit + loading docks
Corridors / Warehouse	Panoramic 360°	Axis M4328-P	1 per 200 m ² open area
Clean Room Entrances	Compact dome	Axis M3116-LVE	Every gowning room entry/exit
Server Room / MDF	Fixed with analytics	Axis P3265-LVE	2 per room (entry + equipment view)

Camera Platform Standard: Axis Communications is recommended as the standard camera vendor across all plants for consistency in management (AXIS Camera Station Pro), analytics ecosystem (ACAP), and cybersecurity features (Axis Edge Vault, signed firmware). Video management integrates to plant security operations center and optionally to AI-based analytics in the edge/cloud compute layer.

7. Human Activity Monitoring

7.1 Scope & Ethical Framework

Human activity monitoring in a cGMP medical device facility serves three purposes: worker safety, compliance documentation (operator actions as part of batch records), and process optimization. All monitoring must comply with GDPR (given the Ireland baseline), local labor laws, and Hollister's corporate ethics policies. No biometric surveillance or individual performance scoring is implemented without explicit works council/employee representative agreement.

7.2 Monitoring Technologies

Capability	Technology	Vendor/Solution	Data Governance
Badge-Based Access	RFID/NFC proximity readers	HID iCLASS SE + Lenel OnGuard	21 CFR Part 11 e-signature; personnel ID tied to batch records
Operator Position Tracking	UWB real-time location (RTLS)	Sewio UWB RTLS	Zone-level granularity only; no individual path tracking stored >24h
Gowning Compliance	AI vision analytics	Axis ACAP + custom ML model	Alert-only; no video recording of individuals; aggregate compliance metrics
Ergonomic Assessment	Wearable IMU sensors (voluntary)	Kinetic Reflex system	Opt-in only; anonymized aggregate data to EHS; individual data deleted <7 days
Operator Actions (Batch Record)	Electronic batch record (EBR) interaction	MES system (e.g., MasterControl)	Full 21 CFR Part 11 audit trail; permanent retention per batch
Training Verification	LMS integration + badge scan	ComplianceWire / Veeva Vault QMS	Training records integrated with operator authorization for equipment

8. IoT, Compute, Storage & Network Infrastructure

8.1 IoT Gateway Architecture

IoT gateways bridge the gap between field-level sensors (Level 0-1) and the IT/OT convergence layer (Level 3-3.5). The gateway strategy follows a tiered approach:

Tier	Gateway	Vendor/Model	Deployment	Function
Tier 1: Station	Compact edge gateway	Advantech UNO-2484G or Moxa UC-8200	1 per station (7 per line)	Protocol conversion (IO-Link, Modbus, HART to OPC-UA/MQTT); local buffering
Tier 2: Line	Edge compute gateway	Dell Edge Gateway 5200 or HPE EL300	1 per manufacturing line	Data aggregation, local analytics, ML inference, historian buffering
Tier 3: Plant	Edge server cluster	See Section 8.2 On-Premise Compute	1 per plant (in server room)	Plant historian, MES, SCADA servers, virtual PLC, AI/ML training

8.2 On-Premise Compute Infrastructure

Given the regulatory requirements for data sovereignty, low-latency control, and operational autonomy during WAN outages, the primary compute infrastructure is on-premise with selective cloud extension.

8.2.1 Server Room Configuration (per Plant)

Function	Hardware	Qty	Specifications	Redundancy
Virtualization Hosts	Dell PowerEdge R760	4	2x Xeon Gold 6430, 512GB RAM, 2x 1.6TB NVMe	VMware vSphere HA cluster (N+1)
Historian/Database	Dell PowerEdge R760	2	2x Xeon Gold 6430, 1TB RAM, 8x 3.84TB SSD RAID10	Active-passive replication
Edge AI/ML Inference	Dell PowerEdge R760xa	2	2x Xeon + 2x NVIDIA L40S GPU, 512GB RAM	Active-active load balanced
Backup/DR	Dell PowerEdge R660	2	1x Xeon Silver 4416+, 256GB RAM, 24x 12TB HDD	Veeam replication to secondary site
SAN Storage	Dell PowerStore 1200T	1	50TB usable all-flash, 100GbE NVMe-oF	Dual controller active-active
Network Core	Cisco Catalyst 9500 stack	2	48x 10GbE + 8x 100GbE uplinks	VSS/StackWise Virtual pair

8.2.2 Cloud Extension (Off-Site Compute)

Cloud services supplement on-premise capabilities for non-real-time functions:

Function	Cloud Provider	Service	Data Classification
Enterprise Data Lake	Microsoft Azure	Azure Data Lake Storage Gen2 + Databricks	Anonymized process data; no PII/PHI
Advanced Analytics / AI Training	Microsoft Azure	Azure Machine Learning + GPU Compute	Aggregated production data; no batch-identifiable data
Digital Twin (Enterprise)	Microsoft Azure	Azure Digital Twins + IoT Hub	Plant/line models; validated simulation data
Disaster Recovery	Microsoft Azure	Azure Site Recovery + Blob Storage	Encrypted replicas; geo-redundant
Supply Chain Integration	AWS or Azure	API Gateway + Integration Platform	EDI, ASN, PO data; encrypted in transit/rest

Cloud Selection Rationale: Microsoft Azure is recommended as the primary cloud platform due to its strong life sciences validation support (GxP qualification for Azure), native integration with Microsoft enterprise tools (Active Directory, Power BI), and EU data residency options (Ireland Azure region for GDPR compliance). AWS serves as secondary for supply chain integration where trading partners require it.

8.3 Network Architecture

8.3.1 Network Segmentation (ISA-95 Aligned)

Network Zone	VLAN Range	Switch Platform	Firewall/DMZ	Purpose
Level 0-1: Field	VLAN 100-199	Stratix 5700 (managed)	None (air-gapped from IT)	PLCs, sensors, actuators, I/O
Level 2: Control	VLAN 200-299	Stratix 5400 (L3)	OT Firewall (Palo Alto OT)	DCS, SCADA, historian, HMI
Level 3: MES/MOM	VLAN 300-399	Cisco Catalyst 9300	OT/IT Firewall	MES, QMS, batch records, scheduling
Level 3.5: DMZ	VLAN 350	Cisco Catalyst 9500	Dual firewall (OT + IT)	OPC-UA aggregation, data diode, protocol break
Level 4: Enterprise	VLAN 400-499	Cisco Catalyst 9500	IT Firewall + WAF	ERP, PLM, email, business apps
Level 5: Cloud/WAN	N/A	Cisco SD-WAN (Viptela)	Next-gen firewall + CASB	Cloud services, inter-site connectivity
Guest/IoT	VLAN 900-999	Cisco Catalyst 9300	Isolated; NAC-controlled	Visitor Wi-Fi, non-critical IoT

8.3.2 Network Bandwidth & Redundancy

- Field to Station Gateway: 100 Mbps Ethernet (DLR ring topology) per device cluster
- Station to Line Gateway: 1 Gbps Ethernet (redundant paths) per station
- Line to Plant Core: 10 Gbps fiber uplinks (dual-homed) per line switch stack
- Plant Core to DMZ/Enterprise: 100 Gbps fiber backbone (redundant)
- Plant to Cloud/WAN: Dual ISP with SD-WAN; 1 Gbps primary, 500 Mbps secondary; encrypted overlay
- Wireless (where needed): Wi-Fi 6E (Cisco Catalyst 9164) for mobile devices; private 5G evaluation for future AGV/AMR

9. Metrology, Calibration & Measurement Assurance

9.1 Metrology Strategy

Every measurement device in the digital footprint—from a simple temperature sensor to a coordinate measuring machine—is subject to a calibration management regime traceable to national/international standards (NIST, NPL, PTB). The metrology system is integral to the cGMP compliance framework and directly supports the IQ/OQ/PQ qualification process.

9.2 Calibration Management System

Recommended Platform: Blue Mountain Regulatory Asset Manager (RAM) or Beamex CMX, integrated with the MES/QMS for automated calibration scheduling, out-of-tolerance alerting, and impact assessment on batch records.

Calibration Tier	Scope	Frequency	Standard	Performer
Tier 1: Reference Standards	Plant reference instruments	Annual (accredited lab)	ISO 17025 accredited cal lab	External (e.g., Trescal, Endress+Hauser)
Tier 2: Working Standards	Metrology lab instruments	Semi-annual	Traceable to Tier 1	In-house metrology team
Tier 3: Production Instruments	In-line sensors, transmitters	Per SOP (risk-based: 3-12 months)	Traceable to Tier 2	Calibration technicians
Tier 4: Automated/Self-Calibrating	Smart sensors with self-diagnostics	Continuous (verification) + periodic full cal	IO-Link/HART self-diagnostics	Automated with human verification

9.3 Measurement Assurance Program

- Statistical Process Control (SPC): All critical measurements monitored with automated SPC charting and out-of-control alerts
- Gauge R&R: Required for all new measurement systems per AIAG MSA 4th Edition or equivalent
- Measurement Uncertainty Budgets: Established for every critical-to-quality (CTQ) measurement per GUM (Guide to the Expression of Uncertainty in Measurement)
- Out-of-Tolerance (OOT) Impact Assessment: Automated batch impact analysis when a calibration fails, identifying all batches produced since last known good calibration
- Digital Calibration Certificates: All calibration data stored electronically with 21 CFR Part 11 compliant electronic signatures

10. Qualification Framework: IQ/OQ/PQ & V&V

10.1 Qualification Strategy

The qualification framework ensures every component of the digital footprint is installed, operates, and performs as intended under cGMP conditions. The approach follows GAMP 5 (Second Edition) and aligns with ISPE Commissioning and Qualification Baseline Guide.

10.2 Qualification Scope per Component Category

Component Category	GAMP Cat.	IQ Scope	OQ Scope	PQ Scope
Sensors/Transmitters	Cat 1 (Std HW)	Installation per spec, wiring, calibration verification	Signal accuracy, response time, alarm functions	Performance over 3 production batches; drift analysis
PLCs	Cat 3-4	Hardware install, firmware version, I/O wiring, network config	Logic execution, safety functions, failover, alarm response	Full production scenario including abnormal conditions
DCS/SCADA	Cat 4-5	Server install, software version, network connectivity	All control functions, alarms, historian data integrity, Part 11	3 consecutive production days; operator workflow validation
MES	Cat 5 (Custom)	Server/client install, database config, interfaces	All business rules, electronic batch record, deviation handling	End-to-end batch processing with intentional deviations
IoT Gateways	Cat 3	Installation, network config, protocol config	Data throughput, latency, store-and-forward, failover	Performance under peak load for sustained production
Network Infrastructure	Cat 1-3	Switch/firewall install, VLAN config, cabling certification	Throughput, latency, failover, segmentation validation	Traffic under production load; failover timing
Compute/Storage	Cat 1-3	Hardware install, OS/hypervisor, storage provisioning	Performance, redundancy, backup/restore, access control	Sustained load testing; disaster recovery exercise
Cameras (Quality)	Cat 3-4	Mounting, alignment, lighting, calibration	Detection accuracy, repeatability, lighting sensitivity	Known defect challenge kit; false positive/negative rates
Environmental Monitoring	Cat 3	Sensor placement per mapping study, calibration	Alarm thresholds, data recording, excursion handling	Environmental challenge testing; worst-case scenarios

10.3 V&V for Systems Engineering

The end-to-end Verification and Validation (V&V) process follows the V-Model, ensuring traceability from user requirements through to operational performance:

1. User Requirements Specification (URS): Business and regulatory requirements defining what the system must do
2. Functional Requirements Specification (FRS): How the system will fulfill each URS requirement
3. Design Specification (DS): Detailed technical design including hardware, software, and configuration specifications
4. Factory Acceptance Testing (FAT): Verification at vendor site before shipment (where applicable)
5. Site Acceptance Testing (SAT): Verification of installation and basic function at the Hollister facility
6. Installation Qualification (IQ): Documented verification that equipment is installed per approved specifications
7. Operational Qualification (OQ): Documented verification that equipment operates per approved specifications across operating ranges
8. Performance Qualification (PQ): Documented verification that the system consistently performs as intended under actual production conditions

Traceability Matrix: A requirements traceability matrix (RTM) links every URS requirement through FRS, DS, and test protocols to test results, ensuring complete V&V coverage with no gaps.

11. Manufacturing on Demand (MOD) Accommodations

11.1 MOD Architecture Requirements

Manufacturing on Demand requires the digital footprint to support rapid product changeover, dynamic batch sizing, configurable process parameters, and real-time demand signal integration. The architecture accommodates MOD through the following mechanisms:

11.2 Recipe-Driven Production

- ISA-88 Recipe Model: All production parameters encoded in hierarchical recipes (master recipe, control recipe, equipment recipe) managed within the DCS batch engine
- Dynamic Recipe Selection: MES-driven recipe download to PLCs based on production schedule, allowing automated changeover between products on the same line
- Parameter Sets: Modular parameter groups that can be mixed and matched for new product variants without new PLC programming
- Golden Batch Profiling: Statistical reference profiles for each product, enabling real-time deviation detection during production

11.3 Rapid Changeover Support

- Digital Work Instructions: Augmented reality (AR) or tablet-based changeover guides with visual verification steps, integrated with the MES
- Changeover Verification: Automated sensor-based verification that changeover steps are complete (e.g., vision confirmation of tooling, sensor confirmation of cleaning completion)
- Changeover Time Tracking: Automated SMED (Single Minute Exchange of Die) metrics with OEE integration
- Pre-Staged Recipes: Predictive recipe staging based on production schedule, pre-loading next product parameters during current run

11.4 Dynamic Resource Allocation

- Flexible Line Assignment: Production scheduling system allocates products to lines based on capability, availability, and efficiency optimization
- Shared Equipment Pool: Inspection stations and packaging lines configurable for multiple product families through recipe-driven setup
- Material Kanban: IoT-enabled material replenishment triggered by real-time consumption data from line sensors
- Labor Allocation: Operator skill matrix integrated with MES for dynamic assignment of qualified personnel to production activities

12. Interim PLM Solution

12.1 Interim PLM Strategy

Hollister requires immediate product lifecycle management capabilities but is not yet ready to commit to a permanent enterprise PLM platform. The interim solution must provide essential PLM functions while maintaining clean data models and migration paths that will support any future PLM selection.

Guiding Principle: The interim PLM serves as a structured data container and workflow engine, not as a permanent system of record. All data structures follow open standards (STEP AP242, PLM XML, or equivalent) to ensure vendor-agnostic extractability.

12.2 Interim PLM Platform Recommendation

12.2.1 Primary Recommendation: Aras Innovator (Open Source Core)

Aras Innovator is recommended as the interim PLM for the following reasons:

- Open-source core with no per-seat licensing, reducing commitment risk
- Resilient, enterprise-grade platform used by major manufacturers (Airbus, Motorola)
- Open data model using SQL Server with well-documented schema, enabling straightforward data extraction/migration
- Built-in configuration management, change control, and document management meeting cGMP requirements
- REST API and web services for integration with MES, ERP, and engineering tools
- Supports FDA 21 CFR Part 11 with electronic signatures and audit trails

12.2.2 Alternative: SharePoint-Based PLM Lite (Lower Capability)

If the organization prefers to leverage existing Microsoft infrastructure, a structured SharePoint Online deployment with Power Automate workflows can provide basic PLM functions: document control, change request workflows, BOM management via linked lists. This option is lower-capability but has near-zero incremental cost and familiar user interface. The risk is data structure limitations that may complicate future migration.

12.3 Interim PLM Functional Scope

PLM Function	Interim Implementation	Migration Readiness
Bill of Materials (BOM)	Aras BOM management with part-item-BOM hierarchy	Export to STEP AP242 or CSV; standard BOM structure

Document Control	Aras document management with revision control, approval workflows	File-based extraction with metadata; standard naming conventions
Change Management	ECR/ECO/ECN workflow in Aras with impact analysis	Change history exported as structured data with full audit trail
Configuration Management	Product effectivity, serialization rules, variant management	Configuration baseline snapshots exportable to any PLM
CAD Data Management	Aras vault with CAD connector (SolidWorks, CATIA, AutoCAD)	CAD files + metadata export; neutral format (STEP, JT) mandatory
Supplier Collaboration	Aras Secure Social with supplier portal for drawing markup	Supplier data in standard format; independent of PLM platform
R&D to Mfg Tech Transfer	Stage-gate workflow with design transfer checklist	Transfer package as structured data package; process-independent
Regulatory Submission Support	Document linking, DHF/DMR/DHR organization	Regulatory dossier structure exportable as XML/PDF package

12.4 PLM-MES Integration (Critical Interface)

The interim PLM must feed validated production specifications to the MES/DCS layer. This interface is critical for cGMP compliance and is designed to be PLM-vendor-agnostic:

- Integration Bus: Apache Kafka or Azure Service Bus as the middleware, with standardized B2MML (ISA-95 XML) message formats
- Master Data: Product definitions, BOM, process specifications published from PLM to a shared Master Data Management (MDM) layer
- Recipe Translation: PLM process specifications translated to ISA-88 recipes through a Recipe Management Service, decoupled from both PLM and DCS
- Change Propagation: ECO approval in PLM triggers change notification to MES, which manages the manufacturing-side impact assessment and implementation

Critical Design Decision: The Recipe Management Service is an independent microservice that sits between PLM and DCS. When the permanent PLM is selected, only the PLM-to-RMS interface changes; the RMS-to-DCS interface and all validated recipes remain untouched.

13. Technology Transfer & Change Control

13.1 R&D to Manufacturing Tech Transfer

The technology transfer process is the highest-risk interface in the product lifecycle for a medical device manufacturer. The digital footprint supports smooth handoffs through:

13.1.1 Design Transfer Package (Digital)

- **Digital Design History File (DHF):** Complete product development documentation managed in interim PLM, including design inputs, outputs, verification, and validation records
- **Process Design Package:** Manufacturing process specifications, equipment requirements, critical process parameters (CPPs), and critical quality attributes (CQAs) encoded in structured data formats
- **Recipe Template:** Pre-validated ISA-88 recipe structure with parameterized CPPs, ready for site-specific equipment mapping
- **Measurement System Specification:** Required measurement capabilities, accuracy requirements, and gauge R&R acceptance criteria for every CTQ measurement
- **Validation Master Plan Template:** Pre-defined IQ/OQ/PQ protocol templates populated with product-specific acceptance criteria from design transfer

13.1.2 Digital Thread Continuity

The architecture maintains a digital thread from R&D design intent through manufacturing execution:

- **Requirements Traceability:** URS/FRS requirements link through design specifications to manufacturing parameters to test results
- **Process Analytical Technology (PAT):** Real-time process monitoring parameters defined during R&D validated as in-process controls during manufacturing
- **Knowledge Management:** Lessons learned, process development reports, and deviation histories captured in structured knowledge base accessible to manufacturing engineering

13.2 Change Control Framework

Change control is the governance mechanism ensuring every modification to the validated state is assessed, approved, implemented, and verified. The digital footprint implements a multi-tier change control system:

Change Tier	Scope	Approval	V&V Required	Example
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Tier 1: Critical	Change to validated system affecting product quality or patient safety	QA Director + Engineering Director + Regulatory	Full re-qualification (IQ/OQ/PQ)	PLC program change affecting process control; new sensor type
Tier 2: Major	Change to validated system with indirect quality impact	QA Manager + Engineering Manager	Partial re-qualification (OQ/PQ typically)	Firmware update; network reconfiguration; HMI screen change
Tier 3: Minor	Change to non-validated system or like-for-like replacement	Engineering Manager	Verification testing only	Like-for-like sensor replacement; switch replacement
Tier 4: Emergency	Unplanned change required for production continuity	Shift Supervisor (retroactive full approval within 48h)	Post-implementation qualification	Emergency PLC patch; failed sensor replacement

13.3 Configuration Management

All system configurations are managed as configuration items (CIs) in a Configuration Management Database (CMDB):

- PLC Programs: Stored in version-controlled repository (Rockwell AssetCentre) with change history, approval records, and rollback capability
- DCS Configuration: Server images, historian configuration, alarm setpoint databases versioned and backed up with automated comparison tools
- Network Configuration: Firewall rules, VLAN assignments, switch configs managed via Cisco DNA Center with change approval integration
- Sensor Configuration: IO-Link parameter sets, HART configuration data backed up to centralized repository
- Server/VM Configuration: Infrastructure-as-Code (Ansible/Terraform) for reproducible server builds; golden images maintained for rapid recovery
- Firmware Registry: Approved firmware versions for every device category maintained in CMDB; firmware updates require change control approval

14. cGMP Compliance Architecture

14.1 Regulatory Mapping

The digital footprint is designed with compliance as an inherent property, not an afterthought. The following table maps key regulatory requirements to architectural features:

Regulation	Key Requirements	Architecture Implementation
FDA 21 CFR Part 11	Electronic records, electronic signatures, audit trails	All systems Level 2+ configured with Part 11 controls: e-signatures, audit trails, access control, system validation
FDA 21 CFR 211	cGMP for finished pharmaceuticals (referenced for devices)	Environmental monitoring, calibration management, batch records, deviation handling, CAPA integration
FDA 21 CFR 820	Quality System Regulation for devices	Design controls (DHF in PLM), production controls (MES), CAPA, complaint handling, document control
EU MDR 2017/745	Medical Device Regulation (EU)	UDI integration, post-market surveillance data collection, technical documentation management in PLM
ISO 13485:2016	Quality management systems for medical devices	Process validation (IQ/OQ/PQ), risk management (ISO 14971 integration), measurement management (ISO 10012)
GAMP 5 (2nd Ed.)	Good Automated Manufacturing Practice	Risk-based approach to computerized system validation; categorization of all systems; validation lifecycle documentation
IEC 62443	Industrial cybersecurity	Zone/conduit model, security levels per zone, patch management, access control, incident response
GDPR	Data protection (EU/Ireland)	Human activity data minimization, consent management, data retention limits, right to erasure compliance

14.2 Data Integrity (ALCOA+ Principles)

All electronic data generated by the digital footprint adheres to ALCOA+ principles:

- **Attributable:** Every data entry linked to the person or system that generated it (Part 11 user ID, system ID)
- **Legible:** Data presented in human-readable format with context (units, timestamps, equipment ID)
- **Contemporaneous:** Data recorded at the time of the activity (real-time historian, no post-hoc entry)
- **Original:** Source data preserved in original format; copies clearly identified (historian as system of record)
- **Accurate:** Calibrated instruments, validated calculations, error-checked data entry
- **+Complete:** No selective data deletion; all data including out-of-spec results retained

- +Consistent: Timestamps synchronized (IEEE 1588 PTP); data formats standardized across systems
- +Enduring: Data retained per regulatory retention schedule (minimum 2 years past device expected life for MDR)
- +Available: Data accessible for review throughout its retention period; migration plan for format obsolescence

15. Cybersecurity Architecture

15.1 Defense-in-Depth Strategy

The cybersecurity architecture follows IEC 62443 zone/conduit model with defense-in-depth.

Key design elements:

- Network Segmentation: Air-gapped Level 0-1 from IT; DMZ with data diode for unidirectional data flow from OT to IT where feasible
- Zero Trust: No implicit trust based on network location; all access authenticated and authorized per least privilege
- Endpoint Protection: Application whitelisting on all OT servers (e.g., CyberArk/Trellix) rather than signature-based AV
- Patch Management: OT-specific patch management with vendor-validated patches applied during maintenance windows
- Monitoring: OT-specific IDS/IPS (Claroity or Nozomi Networks) monitoring all OT network traffic for anomalous behavior
- Incident Response: OT-specific incident response plan with defined playbooks for common scenarios (ransomware, unauthorized access, equipment compromise)
- Backup & Recovery: Immutable backups (air-gapped or WORM) with tested recovery procedures; RPO < 1 hour, RTO < 4 hours for critical systems

15.2 Identity & Access Management

- Centralized Identity: Active Directory for IT/OT with dedicated OT OU and separate privileged access management
- Multi-Factor Authentication: Required for all Level 3+ access and all remote access to OT systems
- Privileged Access: CyberArk or equivalent PAM for all administrative access to OT systems; session recording for audit
- Badge-Based OT Access: RFID badge required for PLC programming terminals and HMI engineering access

16. Scaling Model: Line to Plant to Enterprise

16.1 Horizontal Scaling (Adding Lines)

Each manufacturing line is deployed as a self-contained module. Adding a new line follows this process:

1. Clone baseline configuration (PLC programs, DCS templates, network config) from validated golden template
2. Deploy physical infrastructure (sensors, PLCs, network switches, gateways) per Bill of Materials
3. Configure line-specific parameters (line ID, equipment tags, network addresses)
4. Execute IQ: Verify installation against specification
5. Execute OQ: Verify operation using standard test protocols adapted from baseline
6. Execute PQ: Run production qualification batches
7. Integrate into plant-level DCS, historian, MES

Target: New line deployment in 8-12 weeks from equipment installation to PQ completion, leveraging reusable qualification packages from baseline.

16.2 Vertical Scaling (Plant Replication)

Deploying the digital footprint to a new Hollister plant:

1. Plant Assessment: Evaluate existing infrastructure, regulatory environment, and local requirements
2. Plant Template Deployment: Deploy plant-level infrastructure (server room, network core, security systems) from plant template
3. Line-by-Line Deployment: Deploy manufacturing lines using the baseline line template
4. Plant Integration: Connect to enterprise data lake, supply chain integration bus, and corporate IT services
5. Site Qualification: Plant-level IQ/OQ/PQ covering inter-system integration and facility-wide functions

16.3 Enterprise Integration (Supply Chain)

The supply chain digital footprint connects all plants through:

- Enterprise Data Lake: Aggregated production, quality, and supply chain data from all plants
- Digital Twin (Enterprise): Federated digital twin models for each plant, connected for supply chain simulation
- Supply Chain Control Tower: Real-time visibility across all plants, warehouses, and logistics partners
- Unified Quality Dashboard: Cross-plant quality metrics, CAPA tracking, and regulatory submission readiness

- Master Data Hub: Centralized product master, material master, and equipment master synchronized across all plants

17. Implementation Roadmap

17.1 Phased Deployment Plan

Phase	Duration	Scope	Key Deliverables
Phase 0: Foundation	Months 1-3	Architecture validation, vendor selection, standards development	Approved architecture, vendor contracts, PLC/DCS programming standards, qualification master plan
Phase 1: Pilot Line	Months 4-9	Single manufacturing line at Ballina (S1-S7) fully instrumented	Baseline line deployed, IQ/OQ complete, interim PLM operational, MES integration validated
Phase 2: Pilot PQ & Optimization	Months 10-14	PQ execution, performance optimization, lessons learned	PQ complete, golden template validated, deployment playbook refined, MOD capability demonstrated
Phase 3: Ballina Full Deployment	Months 15-24	Remaining 5-7 lines at Ballina + plant infrastructure	All Ballina lines operational, plant historian, security ops center, full environmental monitoring
Phase 4: Second Plant	Months 18-30	Replicate to second plant (overlapping with Phase 3)	Second plant operational using refined templates; cross-plant analytics operational
Phase 5: Enterprise Rollout	Months 24-48	All remaining plants + supply chain integration	Full enterprise digital footprint; supply chain visibility; permanent PLM selection and migration

17.2 Investment Estimate (Order of Magnitude)

The following provides a rough order of magnitude (ROM) investment estimate for the baseline pilot line and subsequent scaling:

Component	Pilot Line (1 Line)	Full Plant (8 Lines)	Notes
PLCs & Safety Controllers	\$350K - \$500K	\$3.2M - \$4.5M	Includes I/O modules, communication modules, racks
DCS/SCADA Software	\$150K - \$250K	\$400K - \$600K	PlantPAX licenses, historian, HMI servers
Sensors & Instrumentation	\$400K - \$600K	\$3.5M - \$5.0M	All sensor categories per architecture
Camera Systems (All Types)	\$200K - \$350K	\$1.5M - \$2.5M	Machine vision, process, security
IoT Gateways & Edge Compute	\$100K - \$180K	\$800K - \$1.2M	Station, line, and plant-level gateways
Server Room Infrastructure	\$0 (use existing)	\$600K - \$900K	Servers, SAN, network core (plant-level)

Network Infrastructure	\$150K - \$250K	\$800K - \$1.2M	Switches, firewalls, cabling, Wi-Fi
Interim PLM	\$50K - \$100K	\$100K - \$200K	Aras Innovator (open source core + services)
Systems Integration	\$500K - \$800K	\$2.5M - \$4.0M	Engineering, programming, configuration
Qualification (IQ/OQ/PQ)	\$300K - \$500K	\$1.5M - \$2.5M	Protocol development, execution, documentation
TOTAL (ROM)	\$2.2M - \$3.5M	\$14.9M - \$22.6M	Excludes manufacturing equipment itself

18. Appendices

Appendix A: Acronym Reference

Acronym	Definition
BMS	Building Management System
BOM	Bill of Materials
CAPA	Corrective and Preventive Action
CMDB	Configuration Management Database
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CTQ	Critical to Quality
DCS	Distributed Control System
DHF	Design History File
DMR	Device Master Record
DMZ	Demilitarized Zone
DLR	Device Level Ring
EBR	Electronic Batch Record
ECO	Engineering Change Order
EMS	Environmental Monitoring System
FAT	Factory Acceptance Testing
FRS	Functional Requirements Specification
GAMP	Good Automated Manufacturing Practice
HMI	Human-Machine Interface
IQ	Installation Qualification
MES	Manufacturing Execution System
MOD	Manufacturing on Demand
MOM	Manufacturing Operations Management
OEE	Overall Equipment Effectiveness
OQ	Operational Qualification
OT	Operational Technology
PAT	Process Analytical Technology
PLM	Product Lifecycle Management
PQ	Performance Qualification
PTP	Precision Time Protocol
RTLS	Real-Time Location System

SAT	Site Acceptance Testing
SCADA	Supervisory Control and Data Acquisition
SPC	Statistical Process Control
URS	User Requirements Specification
V&V	Verification and Validation

Appendix B: Key Vendor Summary

Domain	Primary Vendor	Product/Platform	Role
PLCs	Rockwell Automation	ControlLogix 5580 / CompactLogix 5380 / GuardLogix	Primary automation platform
DCS/SCADA	Rockwell Automation	PlantPAx 5.0 / FactoryTalk View SE	Supervisory control and monitoring
Process Sensors	Endress+Hauser	Various (iTEMP, Promass, Liquiline)	Standardized process instrumentation
Machine Vision	Cognex	In-Sight 9000 / DataMan 475V	Quality inspection and identification
Precision Measurement	Keyence	LJ-X8000 / LS-9030	Dimensional measurement and profiling
Cameras (Facility)	Axis Communications	P/M/Q series cameras	Process monitoring and security
Vibration/Predictive	Fluke / PCB Piezotronics	3561 FC / 356A45	Machine health monitoring
Environmental Monitoring	Vaisala / TSI	HMT330 / AeroTrak 9310	Clean room environmental monitoring
IoT Gateways	Advantech / Moxa	UNO-2484G / UC-8200	Edge data aggregation
Edge/Server Compute	Dell Technologies	PowerEdge R760 / PowerStore	On-premise compute and storage
Cloud Platform	Microsoft Azure	IoT Hub, Data Lake, ML, Digital Twins	Enterprise analytics and cloud extension
Network (OT)	Rockwell (Stratix/Cisco)	Stratix 5700/5400	Industrial network switching
Network (IT)	Cisco	Catalyst 9300/9500	Enterprise network infrastructure
OT Security	Claroty or Nozomi Networks	Platform for OT visibility	OT-specific threat detection
Interim PLM	Aras	Innovator	Product lifecycle management (interim)

Calibration Mgmt	Beamex / Blue Mountain	CMX / RAM	Metrology and calibration management
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--- End of Document ---

This document is a living architecture specification. It will be updated as the design progresses through Phase 0 detailed engineering and vendor selection. All recommendations are subject to final vendor evaluation, site assessment, and regulatory review.